Mohammad Uzair Khan

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Education

BITS Pilani, Dubai Campus — Dubai, UAE
B.E. Computer Science — *Expected 2027*
- Relevant Coursework: Database Management Systems, Machine Learning, Data Structures & Algorithms, Large Language Models

Bishop Conrad Senior Secondary School | *Senior Secondary (Class XII)* — Bareilly, India | *2023*

Data Analytics Projects

Payer Access Intelligence: BELSOMRA vs peers in Medicare | *Python, SQL (DuckDB), Excel, Power BI* — Report | Code
- Harmonised **2.24M** CMS formulary records across **367** Medicare drug plans and eight strengths, joined January and August 2026 snapshots to enrollment without double counting shared formularies; **20** source-audit checks
- BELSOMRA is listed in plans holding **80.5%** of members (DAYVIGO 17.7%, QUVIVIQ 9.3%); showed its rise from 79.5% was enrollment mix, not new listings, and queued **70 plans (3.2M members)** for payer review; Excel workbook (64 values recalculated in 8 scenarios) and Power BI project (29 DAX measures)

Rx Market Pulse: Medicaid drug spend, 2019–2025 | *Python, SQL (DuckDB), Excel, Power BI, Next.js* — Live | Code
- Built a tested pipeline over **36.1M** public Medicaid rows, a Power BI report (star schema, 28 DAX measures, 6 pages) and a web dashboard; harmonised **1,488** manufacturer name variants with **19** quality checks
- Found spend rose **60% to $116.9B** on flat prescriptions (brands: **14%** of prescriptions, **89%** of spend); live-formula Excel territory model ranks 51 states, checked in Excel through 6 scenarios against Python

Field Force Planner: reps and territories from the NPI registry | *Python, SQL (DuckDB), Excel, Next.js* — Live | Code
- Streamed the **9.9M**-record NPI registry, placed **98.9%** of providers in a county and sized a launch at **1,310 diabetes** and **474 oncology** reps; Excel sizing model checked in Excel on **2,730 cells**
- Aligned 3,143 counties into one connected territory per rep: workload spread **7.5%** against 274% for an alignment by county count; **99.6%** of territories within 25% of target

Drug Price Watch: US pharmacy drug prices | *Python, SQL, Excel, GitHub Actions, Next.js* — Live | Code
- Built a price index over **17.3M** weekly NADAC prices (every year ties to the portal): since 2019 brand prices **+22%**, generics **–53%**; **53%** of brand price steps fall in January and July
- Cross-checked Rx Market Pulse's price effect against a second source ($2.5B vs $2.8B); a weekly GitHub Actions watch found **28 generic sitagliptin packs** priced at **36%** of Januvia's rate

Trial Pipeline Benchmark: MSD vs 11 peers | *Python, REST API (JSON), SQL, Excel, Next.js* — Live | Code
- Pulled **36,612** trials from the ClinicalTrials.gov API and resolved **1,304** sponsor names to 12 companies (**25** checks, **264** tests); MSD has the **second-largest** active pipeline (238 studies), **35%** of Phase 2+ on Keytruda

Also built: CampaignOps *(campaign KPIs, synthetic data)* · PharmaEvidence *(label changes)* · Footnote *(filings retrieval)* — GitHub

Experience

Backend Developer Intern, Amaani (Travel-Tech Startup) — Dubai, UAE
BITS Practice School (PS-1) — *Jun 2025 – Aug 2025*
- Developed and maintained backend services in **Python (Flask)**, building RESTful APIs for core travel-booking features
- Documented API contracts with **Swagger/OpenAPI** and validated endpoints end-to-end using **Postman**; implemented request validation and data models

Research & Publications

Lightweight Deep Learning for Tomato Leaf Disease Detection on Edge Devices — 2025 – 2026
First Author; submitted to IEEE MSN 2026 (under review)
- Trained a MobileNetV2 classifier (10 classes) reaching **97.33% test accuracy** (macro-F1 0.963); compressed it to **5.3 MB** with INT8 quantization and ran it on a Raspberry Pi 4B at about 250 ms per image

Technical Skills

Analytics & BI: Excel (SUMIFS, INDEX/MATCH, MAXIFS, PERCENTRANK), Power BI (star schema, DAX), data transformation and harmonisation, data-quality checks, KPI and variance analysis, price indexes, forecasting
Languages & Data: SQL (DuckDB, MySQL), Python (Pandas, NumPy, scikit-learn), REST APIs, TypeScript, Java, C
Visualisation & Tools: Power BI, Excel charts, accessible web dashboards (Next.js/React), Git, GitHub Actions, pytest, Playwright, Postman
AI / ML: TensorFlow/Keras, edge ML, retrieval (BM25), LLM tool calling (Claude, Gemini)

Certifications

The Ultimate Job Ready Data Science Course – CodeWithHarry (2026)